

What Is Semantic Fidelity

Preserving Meaning in the Age of Artificial Intelligence

SFP 01 | Semantic Fidelity Project

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Abstract

Artificial intelligence increasingly mediates how knowledge is retrieved, compressed, generated, and reused. Standard evaluations can determine whether an output is factually correct or supported by a source, yet still miss whether the relationships that organized the source survive transformation.

This paper develops semantic fidelity as a way to examine what is preserved or lost when information moves across systems, media, or representations. It places the concept within the Reality Drift framework and distinguishes it from accuracy, coherence, semantic drift, and system-level loss of corrective contact.

Core Claim

A representation can remain accurate, coherent, and recognizable while losing important parts of the meaning it was meant to carry.

The Problem of Preserved Form and Altered Meaning

A representation can stay recognizable while still changing what the source meant. A summary might preserve the names, events, and conclusions but strip away the uncertainty that shaped the original argument. The same thing happens when a model gets the facts right but changes their relative weight, or when official language is repeated without the conditions that made it meaningful in the first place.

That makes these failures hard to call simply false. The problem is not always the individual claims. It is the relationship between the representation and what it is supposed to preserve. The surface can remain intact while the meaning underneath shifts.

Definition

Semantic fidelity is the degree to which meaning is preserved as information moves across systems, media, or representations. It describes the integrity of the relationship between symbols and the realities they are intended to represent or reference.

High semantic fidelity preserves the relationships that make information meaningful. These may include intent, context, hierarchy, causality, ambiguity, tone, and reference. The exact dimensions depend on what the representation is meant to do. A legal summary must preserve conditions and exceptions. A medical summary must preserve uncertainty, sequence, and clinical relevance. A personal message may depend more heavily on tone and implied purpose.

Semantic fidelity is therefore neither binary nor identical to textual similarity. A transformed output can use different words and remain faithful. It can also repeat the original wording while shifting its meaning through omission, placement, or reuse in a new context.

Representation and Reference

Every fidelity judgment requires a reference relationship. The evaluator must know what the representation is supposed to remain connected to. That reference may be a source document, an observed condition, a speaker's intended meaning, a body of evidence, or a real-world outcome.

This requirement prevents semantic fidelity from becoming a vague preference for nuance. The central questions are concrete. What is being represented? Which relationships must survive? What can correct the representation if it begins to diverge? When the referent is inaccessible, fidelity becomes harder to establish because the representation can only be compared with other representations.

Accuracy, Coherence, and Fidelity

Accuracy concerns whether individual claims are correct. Source faithfulness concerns whether claims are supported by supplied material. Coherence concerns whether an output holds together internally. Semantic fidelity concerns whether the organizing meaning survives.

These qualities can separate. A fluent summary may be accurate at the sentence level yet flatten the causal structure of its source. A coherent institutional account may accurately reproduce approved records while those records no longer track conditions outside the institution. Neither accuracy nor coherence alone proves continued contact.

How Fidelity Degrades

Fidelity commonly degrades through compression, optimization, and recursive transformation. Compression removes detail so that information can travel. Optimization favors the features that make a representation useful within a system. Recursive transformation makes prior representations the input to later ones.

None of these processes necessarily produces drift. Compression can preserve the relationships relevant to a task, and recursive compression can support learning and abstraction. The danger appears when representations are transformed and circulated faster than their connection to the referent can be tested or restored. Lost structure is then inherited by later outputs, while generation can reconstruct a fluent surface without recovering what disappeared.

Position Within the Reality Drift Framework

Within the Reality Drift framework, semantic fidelity is the thing at risk. Semantic drift describes its gradual loss, while Reality Drift describes the broader condition that can emerge when representations begin displacing the realities they were meant to track.

That weakening can accumulate through recursive compression, especially when simplified representations become easier to optimize and circulate than the realities they describe. The key question is whether anything still has the power to force correction. When representations can be compressed, optimized, distributed, and reused faster than reality can correct them, drift can deepen even while the system continues to function.

Implications for Artificial Intelligence

AI systems are often evaluated at the visible output. Semantic fidelity requires examining the entire path from source and prompt through retrieval, compression, generation, memory, and reuse. A system may succeed at each local task while the chain as a whole weakens the relationship to the original referent.

Fidelity-centered design therefore requires access to sources, preserved provenance, explicit uncertainty, tests for omitted relationships, and correction mechanisms that can alter later behavior. The goal is not to prevent transformation. It is to preserve the structures that the transformation must continue to carry.

Conclusion

Representational failure does not always begin with falsehood. It can begin earlier, when context, hierarchy, causality, or uncertainty disappear while the output remains fluent and usable. Detecting that loss requires looking beyond correctness to the relationship between a representation and what it is meant to preserve.

Coherence is not contact.